Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)

Class Test -III Date: 29 March, 2025 Timing: 8:00 - 11:30 Full mark:[**10**]

Solve all five Qs. In questions Q2, Q3, Q4, and Q5 use integers a, b, c then solve.
Notation: $x \equiv x(t)$, $\dot{x} = \frac{d}{dt}x(t) \equiv Dx(t)$, $i = \sqrt{-1}$

1. **(a)** Given $x = r\sin\theta\sec\phi, y = r\cos\theta\sec\phi, z = r\tan\phi$, express $\frac{\partial}{\partial x} = A(r,\theta,\phi)\frac{\partial}{\partial r} + B(r,\theta,\phi)\frac{\partial}{\partial\theta} + C(r,\theta,\phi)\frac{\partial}{\partial\phi}$ **[1.5]**

   Ans. Since $\frac{\partial}{\partial x} = \frac{\partial r}{\partial x}\frac{\partial}{\partial r} + \frac{\partial\theta}{\partial x}\frac{\partial}{\partial\theta} + \frac{\partial\phi}{\partial x}\frac{\partial}{\partial\phi}$, we have $A = \frac{\partial r}{\partial x}, B = \frac{\partial\theta}{\partial x}, C = \frac{\partial\phi}{\partial x}$

   $(x,y,z) \to (r,\theta,\phi)$: $r = \sqrt{x^2+y^2-z^2}; \tan\theta = \frac{x}{y}; \tan\phi = \frac{z}{r}$

   Using $\frac{\partial}{\partial x}$ in both sides of all eqns: $\frac{\partial r}{\partial x} = \frac{x}{r}$; $\sec^2\theta\frac{\partial\theta}{\partial x} = \frac{1}{y}$; $\sec^2\phi\frac{\partial\phi}{\partial x} = -\frac{z}{r^2}\frac{\partial r}{\partial x}$

   $\implies A = \frac{\partial r}{\partial x} = \sin\theta\sec\phi; B = \frac{\partial\theta}{\partial x} = \frac{1}{y\sec^2\theta} = \frac{1}{r}\cos\theta\cos\phi;$ $C = \frac{\partial\phi}{\partial x} = -\frac{z}{r^2\sec^2\phi}\frac{\partial r}{\partial x} = -\frac{1}{r}\sin\theta\sin\phi.$

   (b) Which of the following statements are true about Levi Civita symbols. (A) $\epsilon_{2j3} = \delta_{j1}$ (B) $\epsilon_{2j3} = \delta_{j2}\delta_{j3} - \delta_{j1}$ (C) $\epsilon_{2j3} = \delta_{2j} - \delta_{j3}$ (D) $\epsilon_{2j3} = -\delta_{j1}$ (E) $\epsilon_{2j3} = 1 - \delta_{j1}$ **[0.5]**

   Ans. (A)for $j = 1$ lhs= -1, rhs =1 [False]
   (B)for $j = 1$ lhs= -1, rhs =0 - 1 = -1, for $j = 2, 3$ lhs=0 rhs = 0 -0 =0 [True]
   (C) for $j = 1$ lhs= -1, but rhs = 0-0=0[False]
   (D) for $j = 1$ lhs= -1=rhs and for $j = 2, 3$ lhs= 0=rhs [True]
   (E)for $j = 1$ lhs= -1, rhs=0 [False]

2. **(a)** Find an analytic function $f(z)$ whose complex part is $ax^2 + bxy + cy^2$. **[0.5]**

   Ans. $f(z) = u(x,y) + iv(x,y)$ and $v(x,y) = ax^2 + bxy + cy^2$. From Cauchy-Riemann condition: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = bx + 2cy$, and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -2ax - by$

   From the first eqn: $u = (bx+2cy)x + g(y)$. This gives $\frac{\partial u}{\partial y} = 2cx + \frac{\partial g(y)}{\partial y}$ which must equate to $-2ax - by$. Also note $\nabla^2 v = 2(a+c) \neq 0$ unless (a+c=0), thus there exists no such u.

   Thus, $\frac{\partial g}{\partial y} = -2(a+c)x - by$. For $g(y)$ to be only a function of y we must have $a = -c$, this can not be satisfied by given a, c. If $a + c$ were 0, we would get $\frac{\partial g(y)}{\partial y} = -by \implies g(y) = by^2/2 +$ a constant, and thus $u(x,y) = (bx + 2cy)x + by^2/2$.

   (b) Use the fact that the particular integral (PI) of the diff eqn $(D-q)x(t) = f(t)$ is $e^{qt}\int_0^t f(t')e^{-qt'}dt'$ to find PI of $(D-a)(D-b)x(t) = e^{ct}$ **[1.5]**

   Ans. Say, $(D-b)x(t) = y(t)$. The equation reduces to $(D-a)y(t) = e^{ct}$, thus from the formula $y(t) = e^{at}\int_0^t f(t')e^{-at'}dt' = e^{at}\int_0^t e^{(c-a)t'}dt = \frac{e^{at}-e^{ct}}{a-c}$.

   Since $y(t) = (D-b)x(t)$ we have a new equation $(D-b)x(t) = \frac{e^{at}-e^{ct}}{a-c}$, which can be solved using the same formula, $x(t) = e^{bt}\int_0^t [\frac{e^{at'}-e^{ct'}}{a-c}]e^{-bt'}dt' = \frac{1}{a-c}[\frac{e^{at}-e^{bt}}{a-b} - \frac{e^{ct}-e^{bt}}{c-b}]$

3. Find the unit vectors which are perpendicular to the tangent line of the curves $\Re(ze^z)$ and $\Im(ze^z)$ at $z = z_0 = a + ib$. Find the unit vectors which are parallel. **[2]**

Ans.

$$\begin{aligned} ze^z &= (x + \ iy)\ e^{x+iy} \\ &= (x + \ iy)\ e^x\ e^{iy} \\ &= (x + \ iy)\ e^x\ (\cos y + \ i \sin y) \\ &= e^x\ ((x \cos y - \ y \sin y) + \ i(x \sin y + \ y \cos y)) \end{aligned}$$

Hence, we get:

$$\Re(ze^z) = e^x\ (x \cos y - \ y \sin y) = u(x, y) \quad \textit{and} \quad \Im(ze^z) = e^x\ (x \sin y + \ y \cos y) = v(x, y).$$

In general $\vec{\nabla} f(x, y)|_{x=a,y=b}$ is perpendicular to to the contour curve $f(x, y) = f(a, b)$ at the point $(x, y) = (a, b)$. Thus $\vec{\nabla} u(x, y)_{x=a,y=b}$ and $\vec{\nabla} v(x, y)_{x=a,y=b}$ are the perpendicular vectors and we must normalize them vector by its length to find the unit vectors. Normalize a vector $\vec{A} \implies \frac{\vec{A}}{\sqrt{\vec{A}.\vec{A}}} \equiv \frac{\vec{A}}{|\vec{A}|}$. Now,

$$\begin{aligned} \vec{\nabla} u(x, y) &= \vec{\nabla}(e^x\ (x \cos y - \ y \sin y)) \\ &= e^x\ ((x + 1) \cos y - \ y \sin y,\ -(x + 1) \sin y - \ y \cos y) \end{aligned}$$

Similarly:

$$\begin{aligned} \vec{\nabla} v(x, y) &= \vec{\nabla}(e^x\ (x \sin y + \ y \cos y)) \\ &= e^x\ ((x + 1) \sin y + \ y \cos y,\ (x + 1) \cos y - \ y \sin y) \end{aligned}$$

The unit vectors at $z = z_0 = a + ib$ can be obtained by setting $(x, y) = (a, b)$ and then normalizing;

$$\text{perpendicular to } u(x, y) \ : \ \frac{1}{N}(\ (a + 1) \cos b - \ b \sin b,\ -(a + 1) \sin b - \ b \cos b)) \tag{1}$$

$$\text{perpendicular to } v(x, y) \ : \ \frac{1}{N}((a + 1) \sin b + \ b \cos b,\ (a + 1) \cos b - \ b \sin b) \tag{2}$$

where $N = \sqrt{(1 + a)^2 + b^2}$.

Note that $\vec{\nabla} u.\vec{\nabla} v = 0$ (can be directely verified of from the fact that both z and e^z are analytic, so does ze^z.) Thus $\vec{\nabla} u$ is perpendicular to u and now it is perpendicular to $\vec{\nabla} v$; thus $\vec{\nabla} v$ must be parallel to u. The unit vectors parallel to the tangent line of $u(x, y), v(x, y)$ at $(x, y) = (a, b)$ are

$$\text{parallel to } u(x, y) \ : \ \frac{1}{N}((a + 1) \sin b + \ b \cos b,\ \ (a + 1) \cos b - \ b \sin b) \tag{3}$$

$$\text{parallel to } v(x, y) \ : \ \frac{1}{N}(\ (a + 1) \cos b - \ b \sin b,\ - (a + 1) \sin b + \ b \cos b) \tag{4}$$

4. Use variable transformation to convert the equations $\{\dot{x} = \sec x \cos y; \dot{y} = \sin x \ \text{cosec} y\}$ to linear form and solve them using matrix method for initial condition $x(0) = a$ and $y(0) = b$ **[2]**

Ans. **(a)** $u = \sin(x), v = \cos(y) \implies \dot{u} = \cos(x)\dot{x}, \dot{v} = -\sin(y)\dot{y}$. Thus, the given differential Eqs $\{\cos(x)\ \dot{x} = \cos(y), \sin(y)\ \dot{y} = \sin(x)\}$ can be written as $\{\dot{u} = v, -\dot{v} = u\}$ with initial conditions $u(0) = \sin(x(0)) = \sin(a) v(0) = \cos(y(0)) = \cos(b)$. The solution is then $\begin{bmatrix} u \\ v \end{bmatrix}_t = e^{tM} \begin{bmatrix} \sin(a) \\ \cos(b) \end{bmatrix}$, where

$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Since $Tr(M) = 0, \det(M) = 1,\ \lambda_\pm = \pm i$,

with eigenvectors $|e_\pm\rangle = \begin{bmatrix} 1 \\ \lambda_\pm \end{bmatrix} = \begin{bmatrix} 1 \\ \pm i \end{bmatrix}$.,

Initial vector in the eigenbasis of M : $\begin{bmatrix} \sin(a) \\ \cos(b) \end{bmatrix} = c_+|e_\pm\rangle + c_-|e_-\rangle = \begin{bmatrix} c_+ + c_- \\ i(c_+ - c_-) \end{bmatrix}$

$\implies c_\pm = \frac{\sin(a) \mp i\cos(b)}{2}$.

Finally $\begin{bmatrix} u \\ v \end{bmatrix}_t = e^{tM} \begin{bmatrix} \sin(a) \\ cos(b) \end{bmatrix} = c_+ e^{it}|e_+\rangle + c_- e^{-it}|e_-\rangle = \begin{bmatrix} \sin a \cos t + \cos b \sin t \\ -\sin a \sin t + \cos a \cos t \end{bmatrix}$

Thus $x(t) = \sin^{-1} u(t) = \sin^{-1}(\sin a \cos t + \cos b \sin t)$.

5. **(a)** Given a matrix $M = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$, compute $A = \sin^2(M)$. What is $\lim_{n\to\infty} A^n$? **[2]**

Ans. $\det(M) = ac$ and $(M) = a + c$ thus eigenvalues are a and c. $\implies D = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix}$.

Writing the eigenvector of a and c side by side $U = \begin{bmatrix} \frac{a-b}{c} & 0 \\ 1 & 1 \end{bmatrix}$. $\implies U^{-1} = \frac{1}{a-c}\begin{bmatrix} b & 0 \\ a-c & 1 \end{bmatrix}$.

Function of a matrix: $f(M) = U \;\; f(D) \;\; U^{-1}$

$$A = \sin^2(M) = U \;\; \sin^2(D) \;\; U^{-1} = \frac{1}{a-c}\begin{bmatrix} \frac{a-b}{c} & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} (\sin a)^2 & 0 \\ 0 & (\sin c)^2 \end{bmatrix}\begin{bmatrix} b & 0 \\ a-c & 1 \end{bmatrix} = \begin{bmatrix} (\sin a)^2 & 0 \\ \frac{b((\sin a)^2 - (\sin c)^2)}{a-c} & (\sin c)^2 \end{bmatrix}$$

$$A^n = \begin{bmatrix} (\sin a)^{2n} & 0 \\ \frac{b(\sin a)^{2n} - (\sin c)^{2n})}{a-c} & (\sin c)^{2n} \end{bmatrix}$$

$\lim_{n\to\infty} A^n = 0$. (For any integer a $\sin a^2 < 1$, and thus $\lim_{n\to\infty} [^2]^n = 0$. This forces all the elements of A^n vanish when $n \to \infty$.